



Algorithmic Notation:

✿ Pseudo Code

✿ Flowchart



Algorithmic Notation: Pseudo Code

Algorithm 2.1: (Largest Element in Array) A nonempty array DATA with N numerical values is given. This algorithm finds the location LOC and the value MAX of the largest element of DATA. The variable K is used as a counter.

- Step 1. [Initialize.] Set $K := 1$, $LOC := 1$ and $MAX := DATA[1]$.
- Step 2. [Increment counter.] Set $K := K + 1$.
- Step 3. [Test counter.] If $K > N$, then:
Write: LOC, MAX, and Exit.
- Step 4. [Compare and update.] If $MAX < DATA[K]$, then:
Set $LOC := K$ and $MAX := DATA[K]$.
- Step 5. [Repeat loop.] Go to Step 2.



Conventions Used in Presenting Algorithm:

Identification No: Each algorithm is assigned an identification no, e.g., Algorithm 4.3

Steps: Each instruction in the algorithm is assigned a step no.

Control: Execution control may be transferred to *step n* from *step i* by the statement go to step n.

Exit: The algorithm is ended by the statement exit.

Comments: Additional information are included in the algorithm by enclosing them in brackets.

Variable Name: Variable name will use capital letters.

Assignment statement: :=

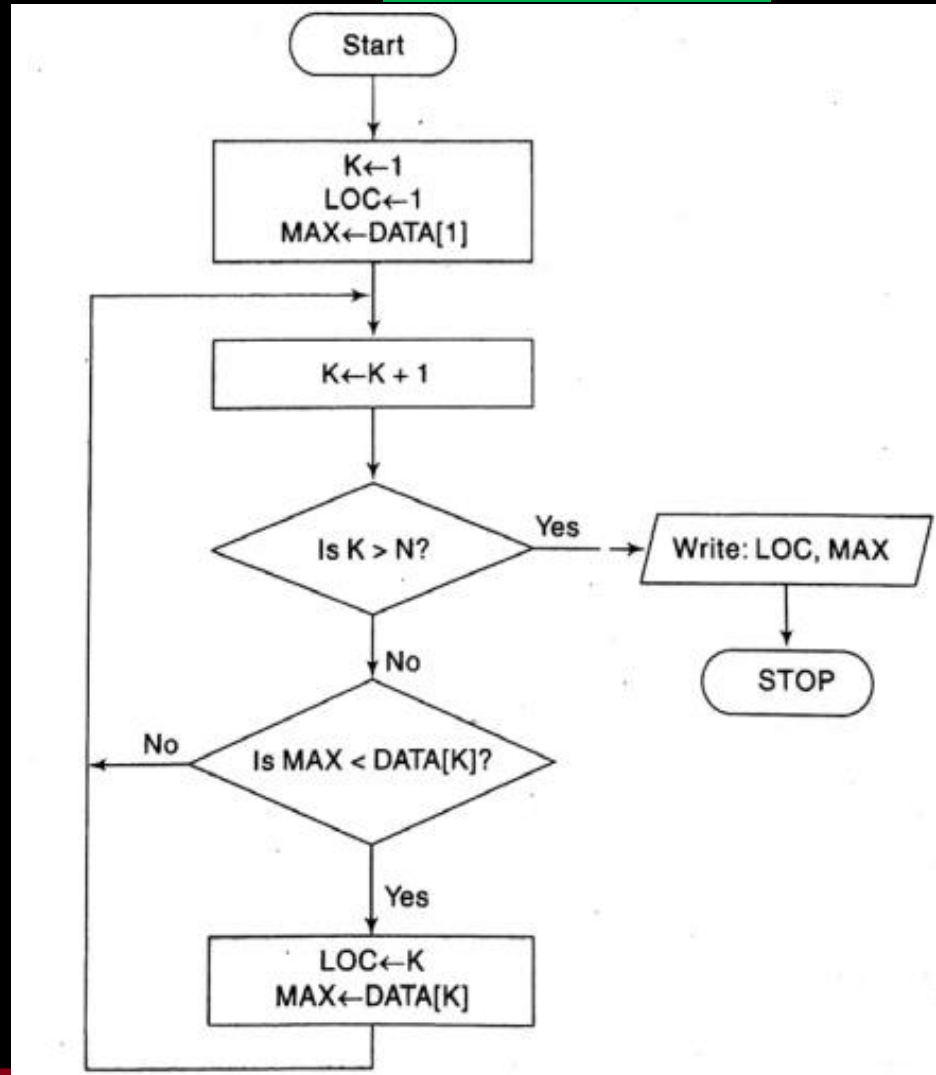
Input: read in variable

Output: write message or variable's value

Procedure: To refer to another independent algorithm.



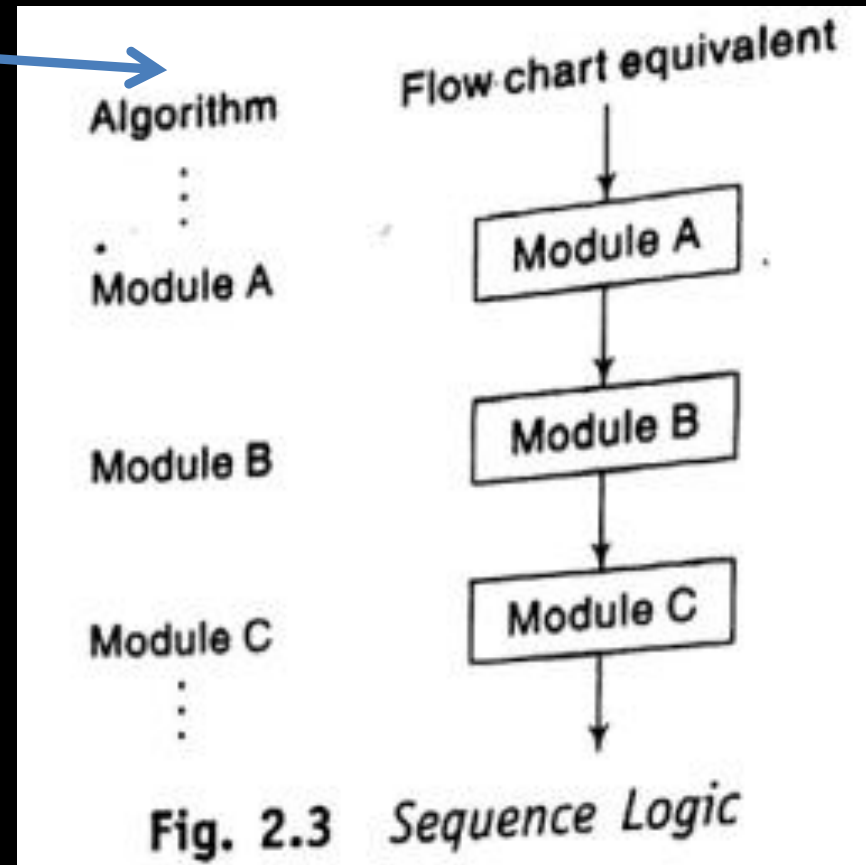
Algorithmic Notation: Flowchart





Control Structure:

- (1) Sequence logic, or sequential flow
- (2) Selection logic, or conditional flow
- (3) Iteration logic, or repetitive flow





Control Structure:

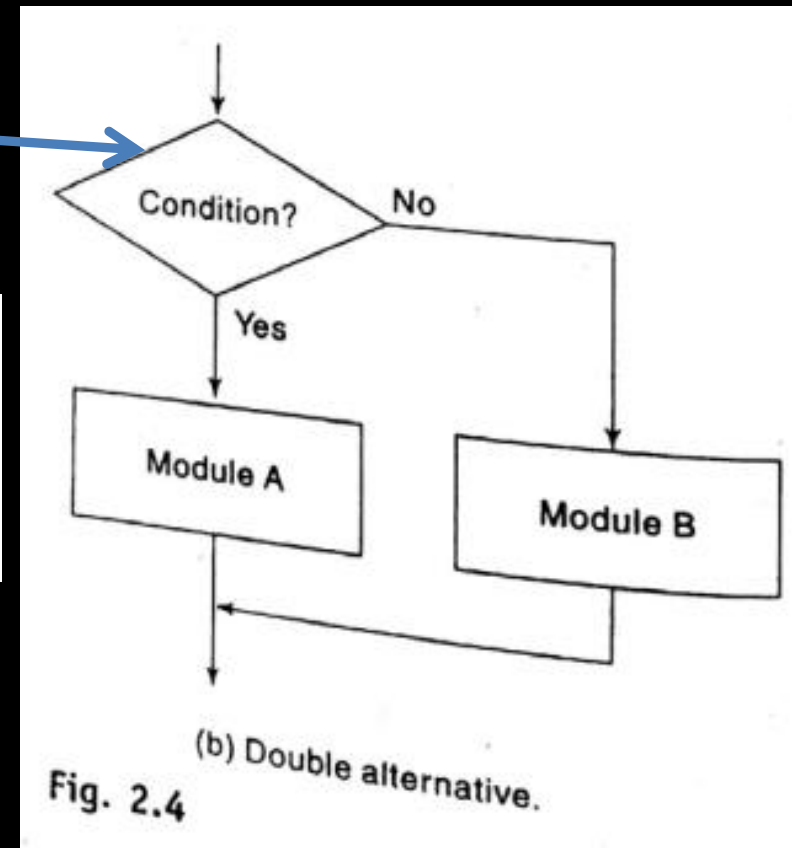
- (1) Sequence logic, or sequential flow
- (2) Selection logic, or conditional flow
- (3) Iteration logic, or repetitive flow

(1) **Single Alternative.** This structure has the form

If condition, then:
[Module A]
[End of If structure.]

(2) **Double Alternative.** This structure has the form

If condition, then:
[Module A]
Else:
[Module B]
[End of If structure.]



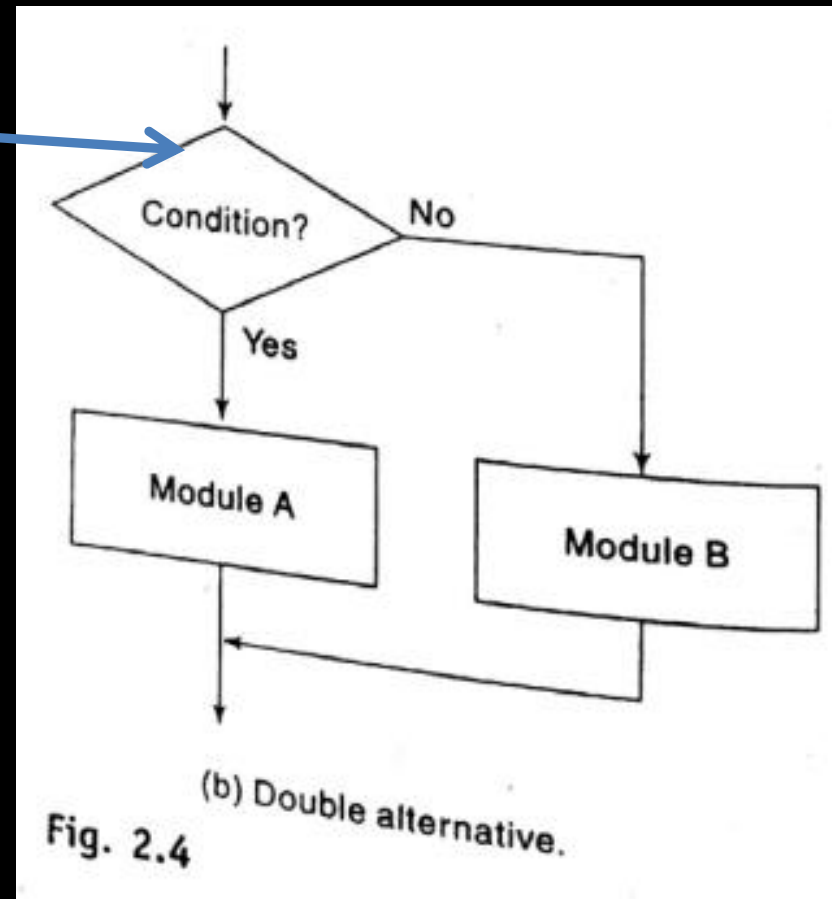


Control Structure:

- (1) Sequence logic, or sequential flow
- (2) Selection logic, or conditional flow
- (3) Iteration logic, or repetitive flow

(3) **Multiple Alternatives.** This structure has the form

If condition(1), then:
 [Module A₁]
Else if condition(2), then:
 [Module A₂]
 :
Else if condition(M), then:
 [Module A_M]
Else:
 [Module B]
[End of If structure.]

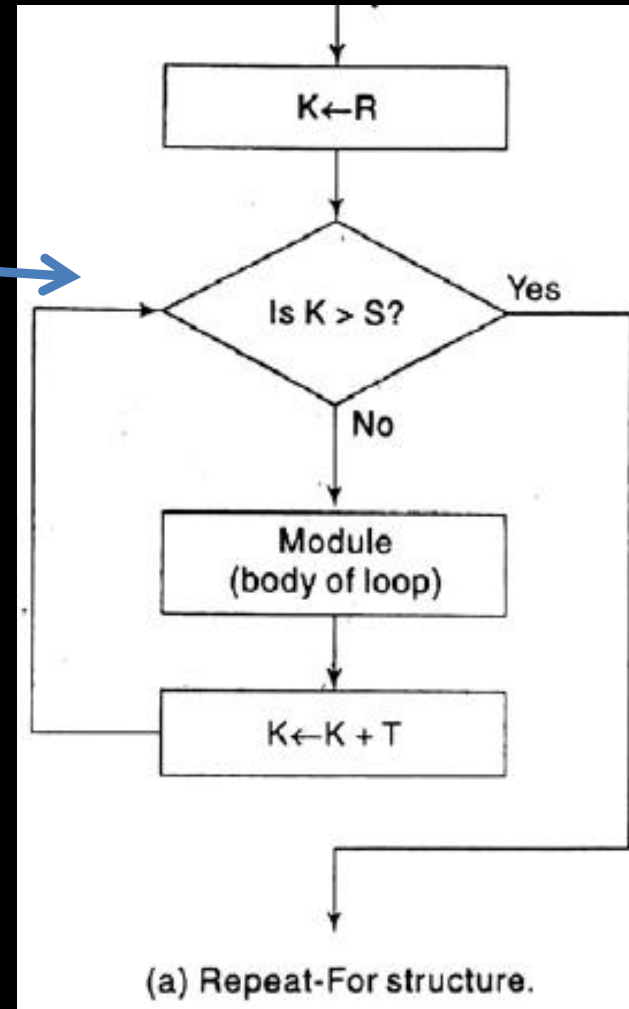




Control Structure:

- (1) Sequence logic, or sequential flow
- (2) Selection logic, or conditional flow
- (3) Iteration logic, or repetitive flow

Repeat for $K = R$ to S by T :
 [Module]
[End of loop.]

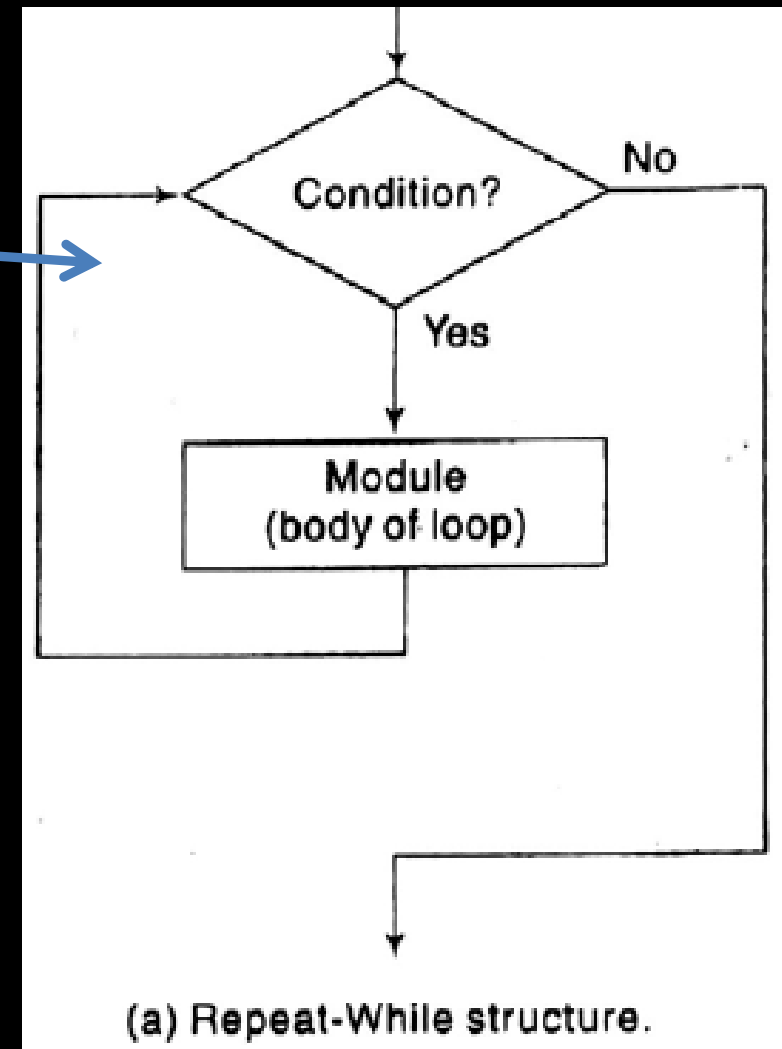




Control Structure:

- (1) Sequence logic, or sequential flow
- (2) Selection logic, or conditional flow
- (3) Iteration logic, or repetitive flow

Repeat while condition:
[Module]
[End of loop.]





Algorithm 2.1 is rewritten using a repeat-while loop rather than a Go to statement:

Algorithm 2.3: (Largest Element in Array) Given a nonempty array DATA with N numerical values, this algorithm finds the location LOC and the value MAX of the largest element of DATA.

1. [Initialize.] Set $K := 1$, $LOC := 1$ and $MAX := DATA[1]$.
2. Repeat Steps 3 and 4 while $K \leq N$;
3. If $MAX < DATA[K]$, then:
 - Set $LOC := K$ and $MAX := DATA[K]$. [End of If structure.]
4. Set $K := K + 1$.
 [End of Step 2 loop.]
5. Write: LOC, MAX.
6. Exit.



2.7 VARIABLES, DATA TYPES

Each variable in any of our algorithms or programs has a data type which determines the code that is used for storing its value. Four such data types follow:

- (1) *Character*. Here data are coded using some character code such as EBCDIC or ASCII. The 8-bit EBCDIC code of some characters appears in Fig. 2.7. A single character is normally stored in a byte.
- (2) *Real (or floating point)*. Here numerical data are coded using the exponential form of the data.
- (3) *Integer (or fixed point)*. Here positive integers are coded using binary representation, and negative integers by some binary variation such as 2's complement.
- (4) *Logical*. Here the variable can have only the value true or false; hence it may be coded using only one bit, 1 for true and 0 for false. (Sometimes the bytes 1111 1111 and 0000 0000 may be used for true and false, respectively.)

The data types of variables in our algorithms will not be explicitly stated as with computer programs but will usually be implied by the context.



CSE 203: Data Structure

Data
Structure



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